



GOVERNEMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSEMENT BOARD
 6th CROSS, MALLESHWARAM, BENGALURU-560003
2026-27 II PUC MODEL QUESTION PAPER- 2

Subject: Chemistry (34)

Maximum Marks: 70

Time: 3.00 Hours

No. of Questions: 46

Instructions

1. Question paper has FIVE parts. All parts are compulsory.
2. a. Part-A carries 20 marks. Each question carries 1 mark.
 b. Part-B carries 06 marks. Each question carries 2 marks.
 c. Part-C carries 15 marks. Each question carries 3 marks.
 d. Part-D carries 20marks. Each question carries 5 marks.
 e. Part-E carries 09 marks. Each question carries 3 marks.
3. In Part-A questions, **first attempted answer** will be considered for awarding marks.
4. Write balanced chemical equations and draw neat labeled diagrams and graphs wherever necessary.
5. Direct answers to the numerical problems without detailed steps and specific unit for final answer will not carry any marks.
6. Use log tables and simple calculator if necessary (use of scientific calculator is not allowed).
7. For a question having circuit diagram/figure/ graph/ diagram, alternate questions are given at the end of question paper in a separate section for visually challenged students.

PART-A

I. Select the correct option from the given choices.

15 × 1 = 15

1. The mixture which shows positive deviation from Raoult's Law is
 (a) n-hexane and n-heptane (b) Bromoethane and chloroethane
 (c) Ethanol and Acetone (d) Chloroform and Acetone
2. A cell that converts energy of combustion of fuels like hydrogen and methane directly into electrical energy is known as
 (a) Dynamo (b) Ni–Cd Cell (c) Fuel Cell (d) Electrolytic Cell
3. The common name of ethanedioic acid is
 (a) Acetic acid (b) Oxalic acid (c) Adipic acid (d) Succinic acid
4. Which aldehyde does not respond to Fehling's test?
 (a) Formaldehyde (b) Ethanal (c) Benzaldehyde (d) Propanal
5. pK_a value of phenol, o-cresol and o-nitrophenol are 10.0, 10.2 and 7.2 respectively, Correct order of acidity is
 (a) Phenol > o-cresol > o-nitrophenol (b) o-cresol > phenol > o-nitrophenol
 (c) o-nitrophenol > o-cresol > phenol (d) o-nitrophenol > phenol > o-cresol
6. A student took two glasses of pure water from a water filter. He cools glass–A in a fridge and warms the other glass – B on a stove. On comparing the solubility of oxygen in H₂O in glass–A and glass–B, he state that glass-‘A’ contains
 (a) more oxygen than glass–B. (b) less oxygen than glass–B.
 (c) same amount of oxygen as in glass–B. (d) zero concentration of oxygen.
7. For a reaction, $2\text{NH}_3(\text{g}) \xrightarrow[\text{Pt-Catalyst}]{1130\text{K}} \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ the rate is
 (a) $\text{Rate} = K[\text{NH}_3]^{\frac{1}{2}}$ (b) $\text{Rate} = K[\text{NH}_3]$ (c) $\text{Rate} = K[\text{NH}_3]^2$ (d) $\text{Rate} = K$

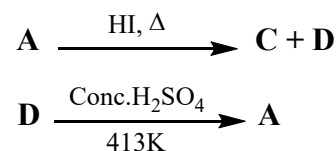
8. Match the following
- | | LIST-I | LIST-II |
|--|----------------------------|---------------------|
| | (i) Phosphodiester Linkage | (A) Amino Acids |
| | (ii) Glycosidic Linkage | (B) Nucleotides |
| | (iii) Peptide Linkage | (C) Monosaccharides |
| | | (D) Phenols |
- (a) i – C, ii – D, iii – B (b) i – C, ii – A, iii – B (c) i – B, ii – C, iii – D (d) i – B, ii – C, iii – A
9. When an alkene reacts with a component 'X' in CCl_4 reddish brown colour of component 'X' disappears to yield vicinal dihalide. Component 'X' is
 (a) Cl_2 (b) HBr (c) Br_2 (d) I_2
10. Statement–I: Potassium permanganate is a powerful oxidant.
 Statement–II: KMnO_4 oxidises I^- to IO_3^- in acidic solution.
 (a) Both Statement I and Statement II are correct.
 (b) Statement I is incorrect but Statement II is correct.
 (c) Both Statement I and Statement II are incorrect.
 (d) Statement I is correct but Statement II is incorrect.
11. Consider the given reaction: $\text{RCONH}_2 + \text{Br}_2 + 4\text{NaOH} \longrightarrow \text{RNH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$;
 If the molecular formula of an amine is CH_5N , then the molecular formula of amide will be
 (a) $\text{C}_2\text{H}_6\text{NO}$ (b) $\text{C}_2\text{H}_5\text{NO}$ (c) CH_5NO (d) $\text{C}_2\text{H}_4\text{NO}$
12. $[\text{Mn}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ are
 (a) Inner orbital and outer orbital complexes respectively. (c) Both inner orbital complexes.
 (b) Outer orbital and inner orbital complexes respectively. (d) Both outer orbital complexes.
13. Triiodomethane was used earlier as an antiseptic. The antiseptic properties are due to
 (a) presence of iodoform. (b) liberation of iodine.
 (c) objectionable smell. (d) presence of methane.
14. Electronic configuration of a transition element X in +3 oxidation state is $[\text{Ar}]3d^5$. What is its atomic number?
 (a) 24 (b) 25 (c) 26 (d) 27
15. Acetylation of salicylic acid produces a compound which has anti-inflammatory and antipyretic property. The compound is
 (a) Bakelite (b) Acetic anhydride (c) Formalin (d) Aspirin
- II. Fill in the blanks by choosing the appropriate word from those given in the brackets:**
- (Primary, Amylose, One, Zero, Picric acid, tertiary) 5 × 1 = 5
16. The sum of mole fractions of all the components in a binary solution is equal to _____.
17. Gabriel Synthesis is used for the preparation of _____ amines.
18. Water soluble component of starch is _____.
19. Oxidation State of 'Ni' in $[\text{Ni}(\text{CO})_4]$ is _____.
20. 2,4,6-trinitrophenol is commonly known as _____.

PART – B

III. Answer any three of the following. Each question carries 2 marks.

3 × 2 = 6

21. Write two differences between order and molecularity of a chemical reaction.
22. Explain Swartz reaction with an example.
23. Give an example for the following. (i) Essential amino acid. (ii) Fibrous protein.
24. Give reason; (i) Separation of Lanthanoids is difficult from their natural occurrence.
(ii) Study of Actinoids is difficult.
25. Compound has a molecular formula $C_4H_{10}O$.
Considering the reactions given below, identify
Compound A and write its IUPAC name.



PART – C

IV. Answer any three of the following. Each question carries 3 marks.

3 × 3 = 9

26. Explain the preparation of potassium permanganate from MnO_2 with balanced equations. Mention the gas evolved when $KMnO_4$ is heated at 513K.
27. On the basis of Valence bond theory, explain the hybridisation, geometry and Magnetic property of complex $[Ni(CN)_4]^{2-}$ [Given Atomic Number of Ni = 28].
28. Write the geometric isomers of complex $[PtCl_2(en)_2]^{2+}$. Which among them is chiral.
29. Considering the elements of 3d transition series, write the element which is
(i) Unable to form a monoxide.
(ii) Having positive $E^0_{(M^{2+}/M)}$ value.
(iii) A base element of Ziegler Catalyst used to manufacture polythene.
30. Alfred Werner a Swiss Chemist formulated his revolutionary ideas about the structure of co-ordination compounds. When a compound $CoCl_3 \cdot 6NH_3$ is treated with excess of $AgNO_3$, 3moles of $AgCl$ are obtained. Based on his observation.
(i) Write the formula of the compound.
(ii) Mention the primary and secondary valency of a central metal atom.

V. Answer any two of the following. Each question carries 3 marks.

2 × 3 = 6

31. What are electrochemical cells? Name the two types of cells.
32. Write three reasons to justify that osmotic pressure method has the advantage over other colligative methods for the measurement of molar mass of the macromolecules.
33. Mention any three factors which decide the products of electrolysis.
34. What is meant by chemical kinetics? Give any two importance of study chemical kinetics.

PART – D

VI. Answer any four of the following. Each question carries 5 marks.

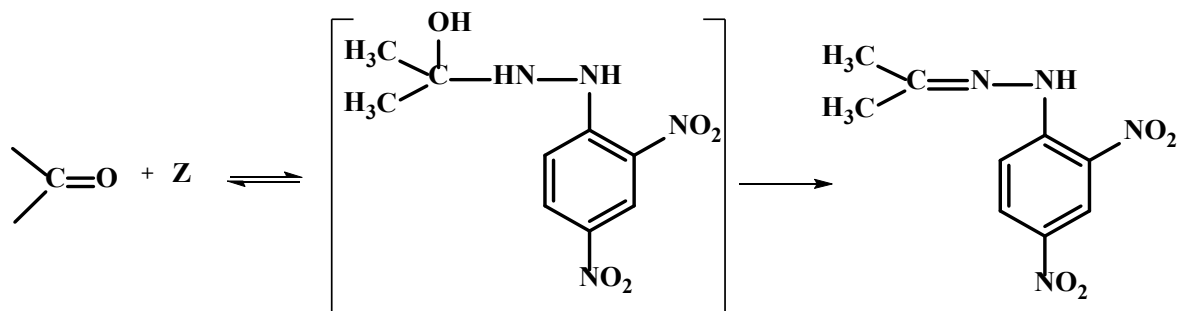
4 × 5 = 20

35. (a) Give the differences between S_N^1 and S_N^2 mechanism with respect to the following.
(i) Order of reaction (Kinetics).
(ii) Configuration of the product.
(iii) Order of reactivity of alkyl halides.
- (b) Explain Wurtz reaction with general chemical equation.

(3+2)

36. (a) Provide the general reaction to prepare Haloform from methyl ketones. Name a reagent which is used to detect the presence of α – methyl group in a carbonyl compound.

(b) In a given chemical reaction, write the structure and name of compound 'Z'. (3+2)



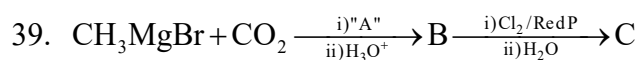
37. (a) How is salicylic acid prepared from phenol? Explain with an equation.

(b) Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Give reason. (3+2)

38. (a) What are reducing sugars? Write the Haworth structure of Lactose.

(b) Hormones are needed in adequate quantities in our body. Give reason.

(c) What is backbone of DNA? (3+1+1)



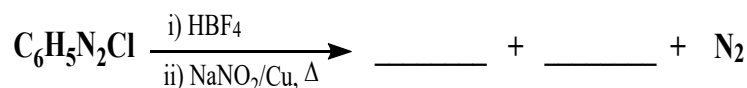
(a) Identify the compound A, B and C in the above reaction.

(b) Name the reaction for the conversion of compound B to C.

(c) Compound 'C' is more acidic than 'B'. Give reason. (3+1+1)

40. (a) Write the chemical reactions to differentiate between primary and secondary amines by using Hinsberg's reagent.

(b) Complete the following reaction:



(c) Write the I.U.P.A.C name of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$.

PART – E (PROBLEMS)

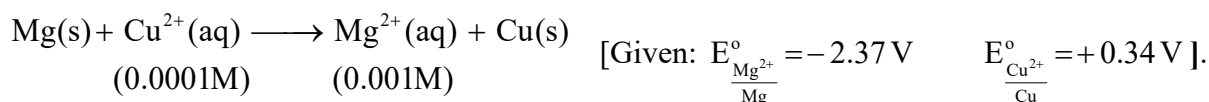
VII. Answer any three of the following. Each question carries 3 marks.

3 × 3 = 9

41. The boiling point of benzene is 353.23K. Calculate the mass of non-volatile solute is to be added to 90g of benzene such that it boils at 354.11K. The molar mass of solute is 58 gmol^{-1} [Given: $K_b = 2.53 \text{ K Kg mol}^{-1}$].

42. Assuming for a very dilute solution calculate the mass of a non-volatile solute which should be dissolved in 114g octane to reduce its vapour pressure to 80%. [Given Molecular formula of octane is C_8H_{18} ; Molar mass of solute = 40 gmol^{-1}].

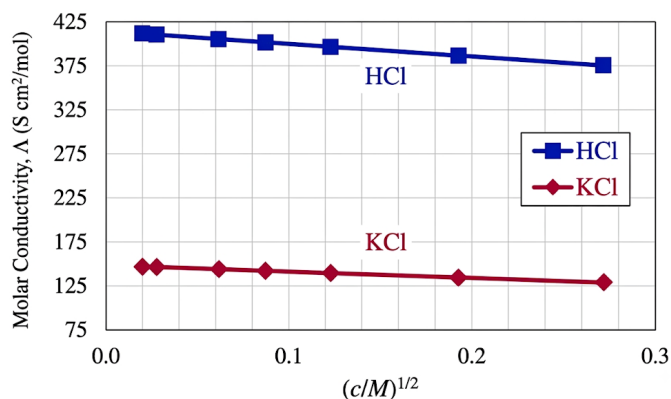
43. Calculate the EMF of the cell for a given reaction.



The rate of a reaction quadruples when the temperature changes from 293K to 313K. Calculate the energy of activation of the reaction assuming that it does not change with temperature.

44. The C-14 content of an ancient piece of wood was found to have $\frac{3}{10}$ of that of living tree. Calculate the age of that piece of wood. [Given that; Half-life of C-14 = 5730 years, $\log 3 = 0.4771$, $\log 7 = 0.8540$].

45. The limiting molar conductivity of CH_3COOH is $390 \text{ Scm}^2/\text{mol}$. Using the graph and given information calculate the limiting molar conductivity of CH_3COOK .



PART – F

(For visually challenged students only)

46. The standard electrode potential for Daniel cell is 1.1V. Calculate the standard Gibbs energy for the reaction. $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ [Given $1F = 96500C$].
